6. Vector Space Models and Embeddings in RAG

1. Introduction to RAG:

→ Essence of RAG is to help to find information and summarize the information.

→ RAG is crucial in modern NLP Applications :-

→ RAG is crucial in modern NLP because it effectively combines the power of (LLM) with the accuracy and freshness of external knowledge sources.

→ Key reasons are →

① Bridging the knowledge gap of LLMs →

→ LLM are trained on large dataset but have a fixed cutoff - they can't access new or domain specific data after training.

→ RAG solve this by letting model retrieve relevant up-to-date information.

② Improved Accuracy and factuality →

→ Traditional LLM can "hallucinate".

→ RAG grounds responses in retrieved, real-world documents, leading to more factually accurate and verifiable answers.

③ Domain Adaptability without retraining →

→ Fine-tuning large models is expensive and time-consuming.

→ RAG allows you to adapt an existing model to a new domain (eg- law, healthcare) simply by changing the retrieval source

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④ Efficient Use of context →

→ LLMs have limited context window

→ Rag intelligently retrieves only the most relevant chunks of information from a large corpus, feeding concise evidence to the model.

⑤ Transparency and Explainability →

→ Since Rag responses are based on retrieved documents, the system can cite sources or show supporting evidence.

→ This transparency builds user trust - crucial in enterprise, academic, and legal ~~academics~~ applications

⑥ Scalability and Maintability →

→ Adding new information is as simple as updating the retrieved database, not retraining the model.

② Representing Text Data Using Embeddings →

→ Some popular Embedding Techniques are →

① Word2vec → captures semantic and syntatic meaning

② Glove → captures global context

③ FastText → handles rare words.

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